

AUTONOMOUS RC CAR WEEK 5 PRESENTATION

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OUR GOAL AS A TEAM

Create an autonomous vehicle that can navigate through an unstructured environment.

Overall Steps

Step 01

Can it run autonomously without using the remote, with sensors and a microcontroller board to which we can upload code?

Step 02

Can it detect and identify obstacles correctly?

Step 03

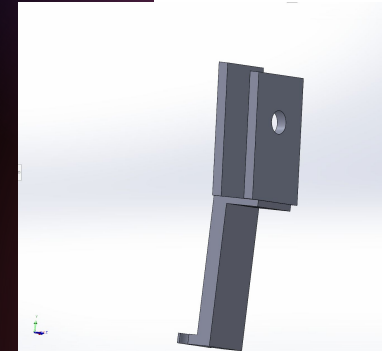
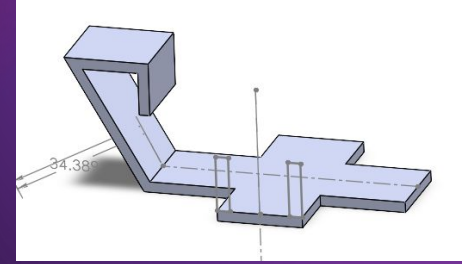
Can it apply that obstacle detection to develop a software that can then avoid the obstacles?

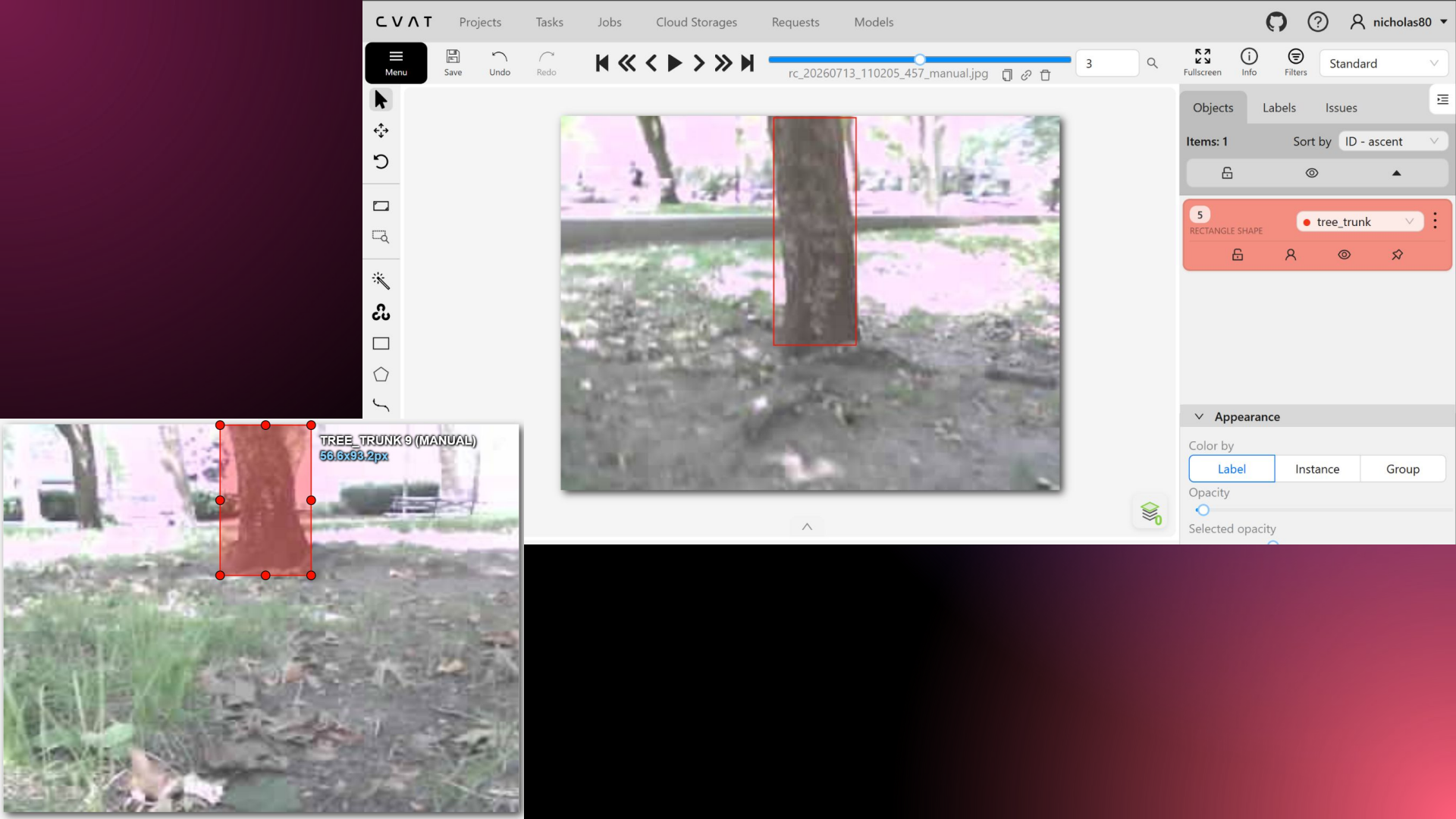
Step 04 (Bonus)

Can it create paths based on given targets?

Progress since our last in person meeting/obstacles encountered

- Started to look into incorporating Lidar sensors into design. Researched and ordered them.
- Designed and printed mounts for the lidar sensors
- Began taking pictures of objects to train AI model
- Stevens-IoT network could not reach park for picture taking
 - Researched and found that we could code the microcontroller to make its own sustainable WiFi network
 - Can now test anywhere, even without being connected to WiFi
- Found out that the camera mount was angled downwards too far, redesigned and reprinted





Demo of Camera System

Next Steps:

- LiDar sensor will be shipped to Nick's house, we will wire it and start writing code to incorporate it into our current obstacle avoidance code that currently uses only the camera
- Testing and training the AI detection, since now the camera mount is angled high enough to take pictures in front of it
- As we develop and test the AI, considering switching the main processor to be onboard, which would be a bigger investment
- Reprinting the LiDar mounts

THANK YOU

Any questions?